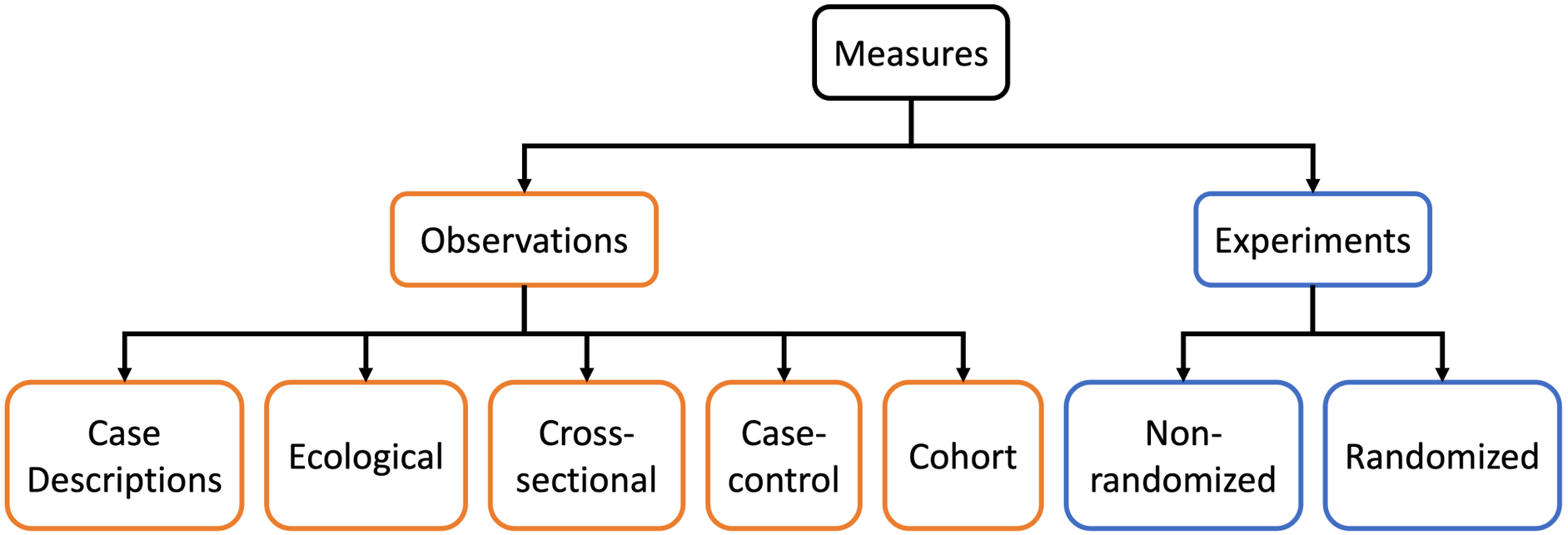
Ecologic Studies and Confounding

APHS 20203 · Epidemiology & Biostatistics

Hierarchy of Study Designs



Typically less convincing → Typically more convincing

Ecologic Studies

- **Ecologic studies compare groups**, such as cities, schools, or countries.
- Researchers analyze **group characteristics**, such as average income and average blood pressure.
- They can describe population patterns and investigate questions about policies or environments.
- A relationship between groups does **not necessarily describe the relationship between individuals**.

Unit of Observation vs. Unit of Analysis

Hypothetical example: Measure residents' income and blood pressure, then compare city averages.

Term	What Does It Mean?	In This Example
Unit of observation	The entity we collect measurements from	Each resident
Unit of analysis	The entity we compare and draw conclusions about	Each city

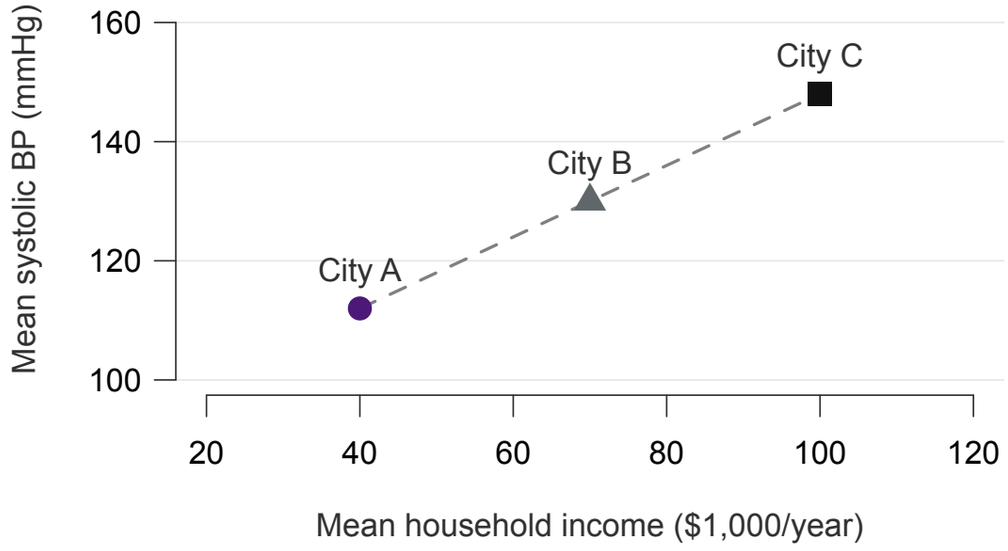
The same individual measurements could support a different analysis that compares **residents** instead of cities.

Ecologic Fallacy

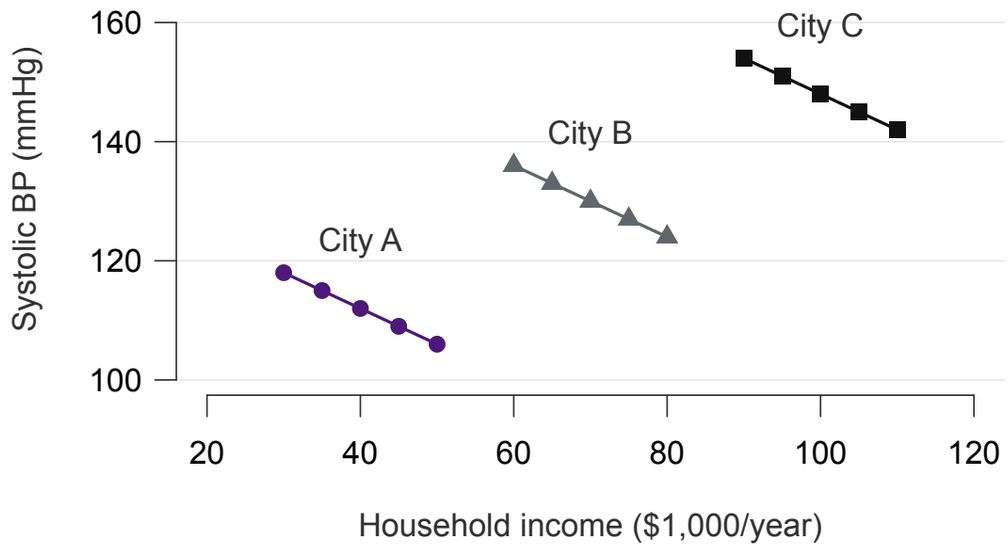
Ecologic fallacy: Assuming a relationship between groups also holds for individuals.

HYPOTHETICAL DATA

Each Point Is a City Average



Each Point Is a Person



City averages alone do not reveal the relationships within cities.

Practice: What Can We Conclude?

Suppose we have **only the city averages** from the previous slide. Which conclusion is supported?

A. Increasing a person's income raises their blood pressure.

B. Within each city, richer residents have higher blood pressure.

C. Cities with higher average income have higher average blood pressure.

D. Income has no relationship with anyone's blood pressure.

When the Population Is the Question

Hypothetical question: After cities adopt a smoke-free indoor-air policy, do their heart-attack admission rates decline compared with cities that do not?

- **Exposure:** A city policy that changes the shared environment.
- **Outcome:** The city's admission rate over time.
- **Why study groups?** We want to evaluate the policy's overall population effect, including effects on people who do not smoke.
- The analysis must still consider other changes occurring in those cities.

Dietary Fat and Breast Cancer Mortality

The question: Could differences in diet help explain why breast cancer death rates differ across countries?

Sasaki et al. (1993)

- **Design:** Ecologic study of 30 countries, using data from 1961 to 1986.
- **Exposure:** National food-supply estimates per person, separated by fat type.
- **Outcome:** Female breast cancer mortality rates.
- **Finding:** Animal fat excluding fish fat was positively associated with mortality among women aged 50 and older. Fish fat showed inverse associations when statistically significant.

The country averages do not tell us what the women who died had eaten.

Low-Fat Dietary Pattern: Randomized Trial

A related question: Does assignment to a dietary program reduce new breast cancers?

Prentice et al. (2006)

- **Participants:** 48,835 postmenopausal women, ages 50–79, without prior breast cancer.
- **Intervention:** A program to reduce total fat and increase fruit, vegetables, and grains; comparison participants were not asked to change their diet.
- **Outcome:** New invasive breast cancers during an average 8.1 years of follow-up.
- **2006 result:** The intervention did not produce a statistically significant reduction in invasive breast cancer incidence.

Ecologic Studies: Strengths and Limitations

Strengths

- Fit questions about population policies and environments
- Reveal patterns across places or time
- Can use existing data efficiently

Limitations

- Group associations may not describe individuals
- Averages can hide differences within groups
- Other population differences can complicate interpretation

Choose the design that fits the question and the level of the conclusion.

Group Discussion: Comparing Study Designs

Work in groups of 2–3 for 4 minutes. Prepare a brief explanation for each question.

	Ecologic Study	Randomized Trial
Unit of analysis	Countries	Individual women
Exposure / intervention	National supply of different fats	Assignment to a dietary program
Outcome	Breast cancer mortality	Invasive breast cancer incidence

1. Why does the ecologic study not tell us whether changing one woman's diet would change her risk?
2. What does random assignment add, and what dietary question does this trial actually test?
3. Why would it be misleading to say that this trial simply proves or disproves the ecologic study?

From Study Design to Trustworthy Comparisons

The two dietary studies asked different questions. For either study, we still need to ask **what could make the comparison misleading**.

- **Selection:** Who entered the study, and who stayed in it?
- **Information:** Were exposure and outcome measured accurately?
- **Confounding:** Did the groups differ in other causes of the outcome?

We will connect these questions to the bias examples from the case-control lecture.

Social Desirability Bias

- **Definition:** Participants report what seems socially acceptable instead of what actually happened.
- **Example:** Someone understates their alcohol use because they worry the interviewer will judge them.
- **Problem:** The recorded answer differs from the behavior we want to measure.
- **Reducing it:** Use neutral questions and protect participants' privacy.

Hawthorne Effect

- **Definition:** People change their behavior because they know they are being observed.
- **Example:** Participants walk more while a research team monitors their activity.
- Their recorded activity may be accurate, but it may differ from their usual activity.

Changing behavior and reporting behavior inaccurately are different problems.

Practice: Bias and Behavior Changes

For each scenario, identify the **bias or behavior change** and explain why.

- Parents of sick children recall pesticide exposure during pregnancy more carefully than parents of healthy children.
- Participants underreport alcohol use because they worry that the interviewer will judge them.
- Outcome assessors know who received treatment and rate those participants more favorably.
- Interviewers ask lung cancer cases more detailed exposure questions than they ask controls.
- Participants exercise more because they know their activity is being monitored.

Review: Sample Size and Bias

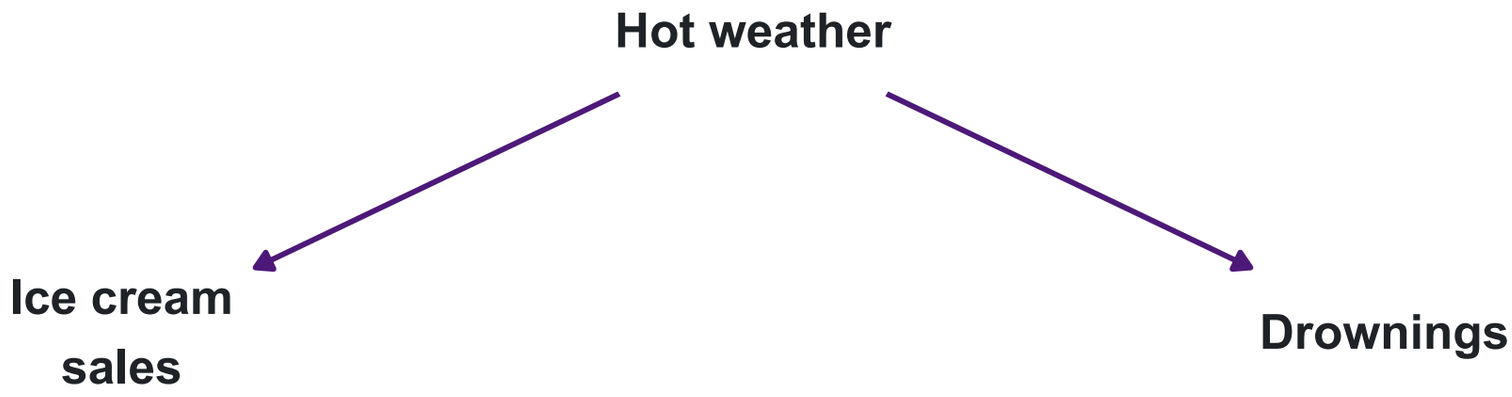
- **Random error (sampling variability):** Results vary by chance. More observations generally improve precision.
- **Bias:** A systematic problem in selection or measurement can remain even in a very large study.

Example: Interviewing more people at local gyms does not fix the problem of estimating community activity from a sample that overrepresents gym users.

For bias, improve **who is included** or **how measurements are made**. Increasing the sample size alone does not repair those processes.

Review: Confounding

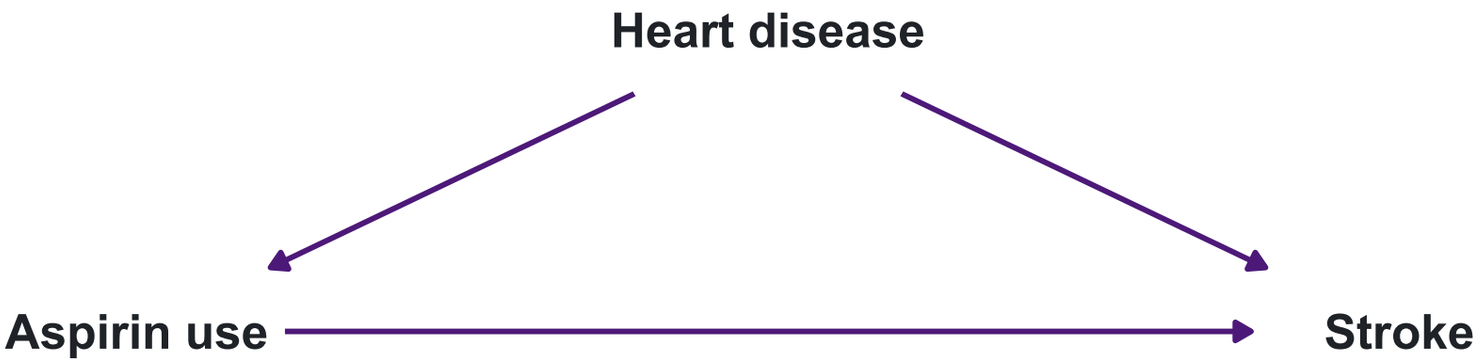
Confounding: Other causes of the outcome get mixed into our estimate of the exposure's effect. We may give credit or blame to the wrong variable.



Would reducing ice cream sales reduce drownings? This pattern alone does not show that it would.

Reading a Causal Diagram

An **arrow** represents an assumed causal influence. It can increase or decrease the outcome.



People taking aspirin may already have a different stroke risk. Comparing users with nonusers can mix that difference with aspirin's effect.

The arrows require knowledge about the problem, not just correlations.

Practice: Does the Class Increase Falls?

Hypothetical study: Follow 200 people for one year after some choose to attend a fall-prevention class.

	People With at Least One Fall	People Followed	One-Year Risk
Attended class	28	100	**28%**
Did not attend	12	100	**12%**

Which conclusion is justified?

A. The class caused more falls.
B. Attendees had more falls, but they may have differed before class.
C. The class prevented falls.

Comparing People With the Same Fall History

The same 200 people, now separated by whether they had fallen **before** the class began:

	No Prior Fall	Prior Fall	All Participants
Attended class	1 / 10 = **10%**	27 / 90 = **30%**	**28 / 100 = 28%**
Did not attend	9 / 90 = **10%**	3 / 10 = **30%**	**12 / 100 = 12%**

- Within each history group, the risks are **the same**.
- Attendees included more people with prior falls: **90% versus 10%**.

Accounting for fall history removes the apparent difference in this example.

Conventional Confounder Criteria

A conventional definition looks for a variable that:

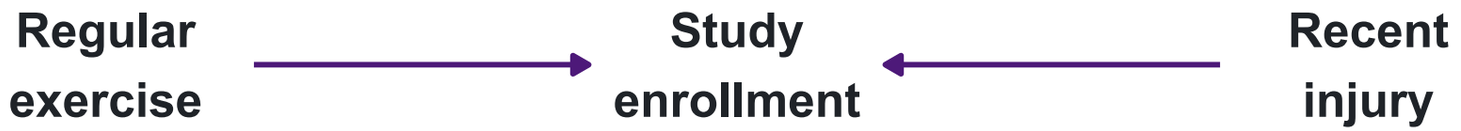
1. Is associated with the exposure.
2. Is associated with the outcome **among people with the same exposure status**.
3. Is not a step through which the exposure causes the outcome.

"Conditional on the exposure" means considering the outcome association separately among exposed and unexposed people.

These criteria are **not a guarantee** that controlling for the variable will help. Its causal role matters. (Hernán et al. 2002)

A Collider: Two Arrows Meet

Hypothetical recruitment rule: Enroll people who exercise regularly **or** have had a recent injury, including those with both.



- Among enrollees, anyone who **does not exercise must have an injury**.
- Exercise could therefore appear protective because of **who was selected**.

A common effect is not a common cause. Controlling for it can create bias.

Practice: Which Variable Needs Attention?

A study compares people who choose a fall-prevention class with people who do not.

Researchers learn that **prior falls influence class attendance and the chance of another fall**.

What is the best reason to consider prior falls when making the comparison?

A. We should control for every variable available.

B. Prior falls are a plausible common cause of attendance and later falls.

C. Any variable associated with the outcome must be controlled.

Addressing Confounding During Study Design

Approach	Basic Idea	Example
Randomization	Assign the intervention by chance	Randomly offer the class
Restriction	Limit a relevant characteristic	Enroll only people with no prior falls
Matching	Select comparison participants with similar characteristics	Match participants on fall history

Randomization balances baseline factors **on average**, including factors we did not measure.

Restriction narrows the population studied. Matching still requires an appropriate analysis.

Addressing Confounding During Analysis

- **Stratification:** Compare exposed and unexposed people within categories of a confounder, as in the fall-history table.
- **Statistical adjustment:** Use methods such as regression or standardization to account for appropriate measured factors.
- Both approaches depend on choosing relevant variables and measuring them adequately.

When reading a study, ask: What did the researchers account for, why did they choose those factors, and what important differences might remain?

Epidemiology Wrap-Up

- Define and explain the **foundations of epidemiology**
- Measure disease using **incidence, prevalence, and rates**
- Describe patterns by **person, place, and time**
- Evaluate **association and causality**
- Distinguish **cohort and case-control** study designs
- Interpret **RR, ARD, and OR**
- Identify major sources of **bias and confounding**

From Concepts to Your Presentation

The Conceptual Progression

- **Descriptive epidemiology:** Describe patterns and generate hypotheses.
- **Analytic epidemiology:** Evaluate exposure–outcome relationships.
- **Critical thinking:** Distinguish association from causation.

For Group Presentation 1

- Describe population patterns using appropriate **epidemiologic measures**.
- Include and cite at least one source of **analytic evidence**.
- Explain the evidence type, exposure, outcome, and main finding.
- Relate the evidence to your findings; discuss **limitations, bias, and confounding**.
- Avoid definitive causal claims or policy recommendations.

Use the **assignment instructions and rubric in D2L** for the full requirements.

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